SCHUR-RANK MULTIPLICATION TENSORS AND STRUCTURAL RIGIDITY OF DETERMINANT ORBIT CLOSURES

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Abstract. We introduce the *Schur-Rank Multiplication Tensor* (SRMT) of a graded G-algebra $A = \mathbf{C}[\mathcal{O}]$:

$$\mathrm{sr}_A^{d_1,d_2}(\lambda,\mu;\nu) := \mathrm{rank}\big(\widetilde{m}_\nu : \mathbf{C}^{N^\nu_{\lambda\mu} c_\lambda(A,d_1) c_\mu(A,d_2)} \to \mathbf{C}^{c_\nu(A,d_1+d_2)}\big),$$

where $\widetilde{m}_\nu$ is the multiplicity-space map induced by the ν-isotypic component of the multiplication map via Schur's lemma (Definition 2.3). We show that the SRMT carries strictly more information than the collection of isotypic multiplicities $\{c_\lambda(A,d)\}$ (Proposition 3.1), and is therefore not obstructed by the Bürgisser–Ikenmeyer–Panova theorem [1].

Let $G = \mathrm{GL}_{m^2}$, $H = \mathrm{GL}_m \times \mathrm{GL}_m$, and let $\mathcal{O}_{\det_m}$, $\mathcal{O}_{\mathrm{perm}_n}$ be the G-orbit closures of the determinant and permanent in $W = \mathrm{Sym}_m(\mathbf{C}^{m^2 *})$. The unconditional results proved in this paper are the following.

(i) **Ray-Multiplicity-1 for the determinant.** The H-weight $k\lambda_{\det}$ occurs with multiplicity exactly 1 in $\mathbf{C}[\mathcal{O}_{\det_m}]_k$ for every $k \geq 0$, where $\lambda_{\det} = (\omega_m;\omega_m)$ is the H-weight of $\det_m$ (Theorem 4.4).

(ii) **Bilinear Rank-1 Rigidity.** For all $a, b \geq 0$ with $a + b \geq 1$: $\mathrm{sr}^{a,b,H}_{\mathcal{O}_{\det_m}}(a\lambda_{\det}, b\lambda_{\det}; (a+b)\lambda_{\det}) = 1$ (Theorem 5.4). The H-isotypic sub-algebra $\bigoplus_{k\geq 0}(\mathbf{C}[\mathcal{O}_{\det_m}]_k)_{k\lambda_{\det}}$ is isomorphic, as a graded ring, to $\mathbf{C}[t]$ with $\deg t = 1$.

(iii) For $n \geq 2$ the diagonal torus $T \cong (\mathbf{C}^*)^{2n-1}$ (the identity component of the stabiliser of perm_n in GL_{n^2} for $n \geq 3$; see Remark 5.8 for $n = 2$) acts by character on perm_n. The one-dimensionality argument used to prove Bilinear Rank-1 Rigidity for the determinant is unavailable for the permanent: $\dim L^H(\mu_{\mathrm{perm}}) = \binom{2n-1}{n}^2 \gg 1$ prevents the dimensional forcing of Theorem 5.4. This does not rule out $\mathrm{sr}(\mathcal{O}_{\mathrm{perm}_n}) = 1$; it means the present method cannot establish it.

(iv) **Non-sphericity of $\mathcal{O}_{\mathrm{perm}_n}$ for $n \geq 3$.** For $n \geq 3$, the orbit $G \cdot \mathrm{perm}_n$ is not spherical for $G = \mathrm{GL}_{n^2}$ (Proposition 8.1). The case $n = 2$ is exceptional and is treated separately in Remark 5.8.

(v) **SRMT Monotonicity.** SRMT values are monotone under orbit-closure inclusion at a fixed degree pair (Theorem 7.1).

Scope disclaimer. These results establish an unconditional structural distinction between $\mathbf{C}[\mathcal{O}_{\det_m}]$ and $\mathbf{C}[\mathcal{O}_{\mathrm{perm}_n}]$ at the level of the SRMT, and, conditionally on three open computational hypotheses (S1)–(S3), a finite non-inclusion $\mathcal{O}_{\mathrm{perm}_3} \not\subseteq \mathcal{O}_{\det_3}$. Neither statement implies the asymptotic separation VP $\neq$ VNP required by Valiant's conjecture.

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1. Introduction

Background. Valiant's conjecture [8] asserts that perm_n is not a polynomial-size linear projection of $\det_m$. In characteristic zero this is equivalent to the strict non-containment of G-orbit closures

$$\mathcal{O}_{\mathrm{perm}_n} \not\subseteq \mathcal{O}_{\det_m} \subset W = \mathrm{Sym}_m(\mathbf{C}^{m^2*}),$$

for m polynomial in n and all n. The Geometric Complexity Theory (GCT) programme of Mulmuley–Sohoni [6, 7] proposes to witness this non-containment via *occurrence obstructions*: dominant weights

λ such that $c_\lambda(\mathbf{C}[\mathcal{O}_{\mathrm{perm}_n}]) > 0$ but $c_\lambda(\mathbf{C}[\mathcal{O}_{\det_m}]) = 0$. The 2019 theorem of Bürgisser, Ikenmeyer, and Panova [1] (hereafter BIP) closes this approach: for m polynomial in n, the multiplicity functions $\lambda \mapsto c_\lambda(\mathbf{C}[\mathcal{O}_{\det_m}])$ and $\lambda \mapsto c_\lambda(\mathbf{C}[\mathcal{O}_{\mathrm{perm}_n}])$ are asymptotically indistinguishable. No occurrence obstruction exists.

The BIP theorem is, however, a theorem about modules, not algebras. It says nothing about the multiplicative structure of the coordinate rings, the ranks of their multiplication maps, or their syzygy modules. The present paper shows that this discarded information carries a provable structural distinction between $\mathbf{C}[\mathcal{O}_{\det_m}]$ and $\mathbf{C}[\mathcal{O}_{\mathrm{perm}_n}]$.

What this paper does and does not prove. The SRMT framework yields:

- An unconditional structural asymmetry: Bilinear Rank-1 Rigidity holds for the determinant and provably cannot hold, via the same mechanism, for the permanent (Theorem 8.4).
- A conditional finite non-inclusion: if open problems (S1)–(S3) are verified computationally (feasible for $n = 3$ via Macaulay2), then $\mathcal{O}_{\mathrm{perm}_3} \not\subseteq \mathcal{O}_{\det_3}$.

This finite non-inclusion, even if established, does *not* imply VP $\neq$ VNP. Valiant's conjecture requires the non-inclusion for *all* n with m polynomial in n — an asymptotic statement. The SRMT programme is best understood as a new invariant tool that bypasses the BIP barrier and provides a structural separation in specific finite-dimensional cases.

Corrections to earlier drafts. We draw attention to five points where the present version corrects errors in earlier drafts.

(1) *False sphericity argument.* An earlier version invoked Brion's theorem to conclude that $\mathbf{C}[\mathcal{O}_{\det_m}]$ is multiplicity-free as an H-module. This was incorrect: $\mathcal{O}_{\det_m}$ is not spherical for H (Remark 4.3). The corrected paper replaces this by Theorem 4.4 (Ray-Multiplicity-1).

(2) *Wrong contrapositive of Brion.* An earlier version incorrectly concluded "not spherical $\Rightarrow$ not multiplicity-free" for the permanent orbit. The correct implication runs the other way (spherical $\Rightarrow$ multiplicity-free; the converse is false in general).

(3) *Erroneous LR coefficient.* An earlier version claimed $N^{2\mu_{\mathrm{perm}}}_{\mu_{\mathrm{perm}},\mu_{\mathrm{perm}}} > 1$ for $m \geq 2$. The correct value is 1; see Remark 2.9.

(4) *Proposition on non-sphericity of* $\mathcal{O}_{\mathrm{perm}_n}$. The original Proposition 8.1 claimed non-sphericity for all $n \geq 2$, which is false for $n = 2$. Proposition 8.1 is now correctly stated for $n \geq 3$. The case $n = 2$ is handled in Remark 5.8.

(5) *Unjustified conclusion of* VP $\neq$ VNP. An earlier draft's §9.2 concluded "VP $\neq$ VNP" from a conditional finite non-inclusion at $n = 3$. This is logically invalid: a single non-inclusion for

fixed n and m does not establish the asymptotic complexity separation required. This conclusion is removed; see §9 for the correct statement.

2. Definitions and Conventions

Throughout, $\mathbf{C}$ denotes the complex numbers; all algebraic groups and varieties are defined over $\mathbf{C}$. Fix integers $m, n \geq 1$ (with $m \geq n$ when comparing the two orbit closures). Set $G = \mathrm{GL}_{m^2}(\mathbf{C})$ and $W = \mathrm{Sym}_m(\mathbf{C}^{m^2*})$. Here $\mathrm{Sym}_m(V)$ denotes the space of homogeneous polynomials of degree m on V^*, i.e. the m-th symmetric power $\mathrm{Sym}^m(V)$; we use the subscript notation throughout when m refers to the matrix size, and the superscript Sym^k for general symmetric powers of modules. The group G acts on W by linear substitution: $(g \cdot f)(x) = f(g^T x)$.

2.1. **Groups and orbit closures. Convention on m and n.** Throughout, m denotes the size of the ambient space ($G = \mathrm{GL}_{m^2}$, $W = \mathrm{Sym}_m(\mathbf{C}^{m^2*})$), and n denotes the size of the permanent (perm_n is an $n \times n$ permanent). We always assume $n \leq m$. In the computational programme of §9, $n = m = 3$. Results involving the Cauchy decomposition of W use m throughout; when perm_n is compared with $\det_m$ for $n < m$, it is understood to be embedded in W via padding.

Definition 2.1 (Orbit closures)**.**

$$\mathcal{O}_{\det_m} := \overline{G \cdot \det_m} \subset W, \qquad \mathcal{O}_{\mathrm{perm}_n} := \overline{G \cdot \mathrm{perm}_n} \subset W,$$

where the closures are taken in the Zariski topology. We maintain the strict distinction between the open orbit $G \cdot \det_m$ and its Zariski closure $\mathcal{O}_{\det_m}$.

Set $H = \mathrm{GL}_m(\mathbf{C}) \times \mathrm{GL}_m(\mathbf{C})$, embedded in G via $(A, B) \cdot X = AXB^T$ for $(A, B) \in H$ and $X \in M_{m \times m}(\mathbf{C})$.

Remark 2.2 (Roles of G and H)*.* The orbit closures are always closures of $G = \mathrm{GL}_{m^2}$-orbits. The H-orbit of $\det_m$ in W is the one-dimensional subspace $\mathbf{C}^* \cdot \det_m$ (by the character identity below). The automorphism group of $\det_m$ as a polynomial is

$$\mathrm{Aut}(\det_m) \cong (\mathrm{GL}_m \times \mathrm{GL}_m)/\mu_m \rtimes \mathbf{Z}/2\mathbf{Z},$$

where $\mu_m = \{(\zeta I_m, \zeta^{-1} I_m) : \zeta^m = 1\}$ and $\mathbf{Z}/2\mathbf{Z}$ corresponds to transposition $X \mapsto X^T$. The identity component has dimension $2m^2 - 1$.

2.2. **Representations and multiplicities.** Let $\Lambda^+ = \Lambda^+(G)$ denote the dominant integral weights of G. For a rational G-module V and $\lambda \in \Lambda^+$:

$$c_\lambda(V) := \dim \mathrm{Hom}_G(L(\lambda), V).$$

For a graded G-algebra $A = \bigoplus_{d\geq 0} A_d$: $c_\lambda(A,d) := c_\lambda(A_d)$. The Littlewood–Richardson coefficient is $N^\nu_{\lambda\mu} := \dim \mathrm{Hom}_G(L(\nu), L(\lambda)\otimes L(\mu))$. The weight semigroup of an orbit closure $\mathcal{O}$ is $\mathcal{X}(\mathcal{O}) := \{\lambda \in \Lambda^+ \mid c_\lambda(\mathbf{C}[\mathcal{O}]) > 0\}$.

2.3. **Formal definition of the SRMT.** The following definition requires careful parsing. For $\lambda \in \Lambda^+$ and a graded G-module M, the λ-*isotypic component* of M is $(M)_\lambda \cong L(\lambda)^{\oplus c_\lambda(M)}$ as a G-module.

The multiplication map $m : A_{d_1} \otimes A_{d_2} \to A_{d_1+d_2}$ is G-equivariant. Its ν-isotypic component is the induced map

$$m_\nu : (A_{d_1} \otimes A_{d_2})_\nu \to (A_{d_1+d_2})_\nu.$$

By Schur's lemma, the ν-isotypic part of $(A_{d_1})_\lambda \otimes (A_{d_2})_\mu$ is $L(\nu)^{\oplus N^\nu_{\lambda\mu} c_\lambda(A,d_1) c_\mu(A,d_2)}$. Any G-equivariant map $L(\nu)^{\oplus p} \to L(\nu)^{\oplus q}$ is determined by a linear map of multiplicity spaces $\mathbf{C}^p \to \mathbf{C}^q$.

Definition 2.3 (SRMT and SRMT defect)**.** Let $A = \bigoplus_{d\geq 0} A_d$ be a connected graded G-algebra and let $d_1, d_2 \geq 0$. For $\lambda, \mu, \nu \in \Lambda^+$ define the degree-(d_1, d_2) *Schur-Rank Multiplication Tensor*:

$$\mathrm{sr}_A^{d_1,d_2}(\lambda,\mu;\nu) := \mathrm{rank}\big(\widetilde{m}_\nu : \mathbf{C}^{N^\nu_{\lambda\mu} c_\lambda(A,d_1) c_\mu(A,d_2)} \to \mathbf{C}^{c_\nu(A,d_1+d_2)}\big),$$

where $\widetilde{m}_\nu$ is the multiplicity-space map induced by m_ν via Schur's lemma, uniquely defined up to choice of bases in the multiplicity spaces; the rank is independent of this choice.

Equivalently, at the level of the actual linear map:

$$\mathrm{sr}_A^{d_1,d_2}(\lambda,\mu;\nu) = \frac{\mathrm{rank}\big(m_\nu\big|_{(A_{d_1})_\lambda \otimes (A_{d_2})_\mu}\big)}{\dim L(\nu)}.$$

This formula is valid because: (i) by G-equivariance, the map $m_\nu|_{(A_{d_1})_\lambda\otimes(A_{d_2})_\mu}$ acts only on the ν-isotypic part of the domain; (ii) by Schur's lemma, the restriction to this part factors as $(\widetilde{m}_\nu \otimes \mathrm{id}_{L(\nu)})$; (iii) the rank of the actual map therefore equals $\mathrm{rank}(\widetilde{m}_\nu)\cdot \dim L(\nu)$. Dividing by $\dim L(\nu)$ (a positive integer) yields $\mathrm{rank}(\widetilde{m}_\nu) = \mathrm{sr}_A^{d_1,d_2}(\lambda,\mu;\nu) \in \mathbf{Z}_{\geq 0}$. In the multiplicity-free case ($c_\lambda, c_\mu, c_\nu \leq 1$) this simplifies to 0 or 1.

The *SRMT defect* is:

$$\delta_A^{d_1,d_2}(\lambda,\mu;\nu) := \min\big(N^\nu_{\lambda\mu} c_\lambda(A,d_1) c_\mu(A,d_2),\ c_\nu(A,d_1+d_2)\big) - \mathrm{sr}_A^{d_1,d_2}(\lambda,\mu;\nu) \geq 0.$$

When d_1 (resp. d_2) is the smallest degree in which $c_\lambda(A,d_1) > 0$ (resp. $c_\mu(A,d_2) > 0$), we drop the superscript and write $\mathrm{sr}_A(\lambda,\mu;\nu)$.

Remark 2.4 (The SRMT and the BIP barrier). The SRMT is specifically designed to operate in the regime where BIP provides no information. Even when $c_\mu(\mathrm{perm}) = c_\mu(\det)$, the ranks of the isotypic multiplication maps can differ between the two algebras. This rank information is invisible to any module-theoretic analysis.

Definition 2.5 (SRMT Monotonicity — informal statement)**.** If $\mathcal{O}_1 \subseteq \mathcal{O}_2$ are G-orbit closures and $d_1, d_2 \geq 0$ are fixed degrees, the Zariski-closed inclusion induces

$$\mathrm{sr}^{d_1,d_2}_{\mathcal{O}_1}(\lambda,\mu;\nu) \leq \mathrm{sr}^{d_1,d_2}_{\mathcal{O}_2}(\lambda,\mu;\nu) \quad \text{for all } (\lambda,\mu,\nu).$$

See Theorem 7.1 and Appendix C for the full proof.

2.4. Cauchy decomposition and the H-module structure of W.

Proposition 2.6 (Cauchy decomposition)**.** *Let $U = V = \mathbf{C}^m$, so that $\mathbf{C}^{m^2} \cong U \otimes V$. The following hold as $H = \mathrm{GL}(U) \times \mathrm{GL}(V)$-module isomorphisms.*

(i) (Cauchy identity.) $W = \mathrm{Sym}_m(U^* \otimes V^*) \cong \bigoplus_{\lambda \vdash m} S_\lambda(U^*) \otimes S_\lambda(V^*)$, where the sum runs over all partitions λ of m. In particular, W is a multiplicity-free H-module.

(ii) (Determinant.) $\det_m$ spans the unique H-component $\lambda = (1^m)$: $S_{(1^m)}(U^*) \otimes S_{(1^m)}(V^*) = \wedge^m U^* \otimes \wedge^m V^*$, of dimension 1 and H-highest weight $\lambda_{\det} = (\omega_m; \omega_m)$.

(iii) (Permanent.) In the case $n = m$, perm_m spans the unique H-component $\lambda = (m)$: $\mathrm{Sym}_m(U^*) \otimes \mathrm{Sym}_m(V^*)$, with H-highest weight $\mu_{\mathrm{perm}} = (m\omega_1; m\omega_1)$ and dimension $\binom{2m-1}{m}^2$. For general $n \leq m$, perm_n is embedded into W via padding and lies in the corresponding H-isotypic component.

(iv) (Global G-structure.) As a $G = \mathrm{GL}_{m^2}$-module, W is irreducible of highest weight $m\omega_1^{\mathrm{GL}_{m^2}}$. Hence $\det_m$ and perm_n belong to the same GL_{m^2}-module; the distinction is an H-module phenomenon.

Remark 2.7 (Torus weight vs. Schur module label)*.* Both $\det_m$ and perm_n are eigenvectors for the diagonal torus $T_H \subset H$ with identical torus character $(\omega_m; \omega_m) = (1, \ldots, 1; 1, \ldots, 1)$. However, they live in different H-irreducible components:

$$\det_m \in S_{(1^m)}(U^*) \otimes S_{(1^m)}(V^*), \quad \lambda_{\det} = (\omega_m; \omega_m), \quad \dim L^H(\lambda_{\det}) = 1;$$

$$\mathrm{perm}_n \in \mathrm{Sym}_m(U^*) \otimes \mathrm{Sym}_m(V^*), \quad \mu_{\mathrm{perm}} = (m\omega_1; m\omega_1), \quad \dim L^H(\mu_{\mathrm{perm}}) = \binom{2m-1}{m}^2.$$

All multiplicity computations and SRMT analysis operate at the level of H-highest weights, not torus characters.

Remark 2.8 (Notation for coordinate-ring elements)*.* We systematically distinguish:

- $\det_m \in W$: the determinant polynomial, a vector in W. Under H, $\det_m(A^T X B) = \det(A)\det(B)\det_m(X)$, so it is an H-eigenvector of weight $\lambda_{\det}$.
- $\widehat{\det}_m \in A_1 = (\mathbf{C}[\mathcal{O}_{\det_m}])_1$: the image of the unique (up to scalar) H-eigenvector of weight $\lambda_{\det}$ in $W^* \cong \mathbf{C}[W]_1$ under the restriction map $\mathbf{C}[W]_1 \twoheadrightarrow A_1$.

Throughout all proofs, we use $\widehat{\det}_m$ (resp. $\widehat{\det}_m^{\,k}$) exclusively to denote elements of the coordinate ring $A_k = (\mathbf{C}[\mathcal{O}_{\det_m}])_k$. We write $\det_m$ only for the polynomial vector in W. This convention is enforced uniformly from Lemma 5.2 onward.

Remark 2.9 (LR coefficients for H). For $H = \mathrm{GL}_m \times \mathrm{GL}_m$, the tensor product rule factorises: $L^H(\alpha;\beta)\otimes L^H(\alpha';\beta') \cong \bigoplus_{\gamma,\delta}(N^{\gamma}_{\alpha\alpha'}N^{\delta}_{\beta\beta'})L^H(\gamma;\delta)$.

For the determinant weight. Since $\lambda_{\det} = (\omega_m;\omega_m)$ is the determinant character of H, the representation $L^H(\lambda_{\det})$ is one-dimensional: $L^H(\lambda_{\det}) = \wedge^m\mathbf{C}^m \otimes \wedge^m\mathbf{C}^m$. The representation $L^H(k\lambda_{\det})$ is then the k-fold tensor product of a one-dimensional character, hence one-dimensional for every $k \geq 0$. Consequently,

$$N^{(a+b)\lambda_{\det}}_{a\lambda_{\det},\, b\lambda_{\det}} = 1 \quad \text{for all } a, b \geq 0,$$

since the tensor product of two one-dimensional representations is one-dimensional; no Pieri computation is needed.

For the permanent weight. The Pieri rule for GL_m gives, for the tensor product $L(m) \otimes L(m)$:

$$L(m) \otimes L(m) \cong \bigoplus_{j=0}^{m} L(2m-j, j),$$

where the sum runs over $j = 0, 1, \ldots, m$ subject to the resulting partition $(2m-j, j)$ fitting in an m-row diagram, and each LR coefficient $N^{(2m-j,j)}_{(m),(m)} = 1$. In particular, there are *multiple* dominant weights in $L(m) \otimes L(m)$, not just $(2m)$. The specific coefficient needed is $N^{(2m)}_{(m),(m)} = 1$: the unique horizontal strip of size m from (m) to $(2m)$ consists of all m boxes placed in row 1. Therefore:

$$N^{2\mu_{\mathrm{perm}}}_{\mu_{\mathrm{perm}},\mu_{\mathrm{perm}}} = \big(N^{(2m)}_{(m),(m)}\big)^2 = 1.$$

> **Warning.** A previous draft claimed $N^{2\mu_{\mathrm{perm}}}_{\mu_{\mathrm{perm}},\mu_{\mathrm{perm}}} > 1$ for $m \geq 2$, and stated that "the only partition of $2m$ obtainable from (m) by adding m boxes with no two in the same column is $(2m)$." Both statements are incorrect. The Pieri rule yields partitions $(2m-j, j)$ for $j = 0, 1, \ldots, m$; there are multiple obtainable partitions. The correct value of $N^{2\mu_{\mathrm{perm}}}_{\mu_{\mathrm{perm}},\mu_{\mathrm{perm}}}$ is 1, justified by the unique horizontal strip to $(2m)$ as above.

3. The Module–Algebra Gap

Proposition 3.1 (Module data does not determine SRMT)**.** *There exist pairs of connected graded G-algebras A, B with $c_\lambda(A,d) = c_\lambda(B,d)$ for all $\lambda \in \Lambda^+$ and $d \geq 0$, yet $\mathrm{sr}_A(\lambda_0,\mu_0;\nu_0) \neq \mathrm{sr}_B(\lambda_0,\mu_0;\nu_0)$ for some triple (λ_0,μ_0,ν_0).*

Scope note: *The algebras A and B constructed in the proof are not coordinate rings of orbit closures. This proposition establishes that, at the level of abstract graded G-algebras, the SRMT carries strictly more information than the multiplicity table, so BIP provides no obstruction to SRMT-based separation arguments in that generality. Whether this holds for orbit-closure coordinate rings is Open Problem (S4) (§9); the GCT separation strategy of §9 does not require (S4).*

Proof. Let $G = \mathrm{GL}_2$ and $M_1 = L(\omega_1)^{\oplus 2}$. Let M_2 be a fixed G-module containing $L(1,1)$ with multiplicity 1 (e.g., $M_2 = L(2,0) \oplus L(1,1)$). Define graded G-algebras:

$$A_0 = B_0 = \mathbf{C}, \quad A_1 = B_1 = M_1, \quad A_2 = B_2 = M_2, \quad A_d = B_d = 0 \text{ for } d \geq 3.$$

Define multiplication maps $m^A, m^B : \mathrm{Sym}^2(M_1) \to M_2$ as follows: both act as the identity on the $L(2,0)$-summand; m^A acts as the identity on the $L(1,1)$-summand, while m^B acts as zero on the $L(1,1)$-summand. (By Schur's lemma, these are the only possibilities up to scalar.)

Equal multiplicities. By construction, $(A_d)_\mu = (B_d)_\mu$ as vector spaces for all (μ, d).

Associativity. Since $A_d = B_d = 0$ for $d \geq 3$, every associativity relation is of the form $f \cdot (g \cdot h) = (f \cdot g) \cdot h$ with total degree ≥ 3, which lies in $A_3 = 0$ and holds trivially. (Both A and B are quotients of the free graded algebra $T(M_1)$ by the ideal of all degree-≥ 3 elements, and both are valid commutative graded algebras.)

Distinct SRMT values.

$$\mathrm{sr}_A(L(\omega_1), L(\omega_1); L(1,1)) = 1 \neq 0 = \mathrm{sr}_B(L(\omega_1), L(\omega_1); L(1,1)). \quad \square$$

4. Multiplicity-Free Algebras, Spherical Varieties, and $\{0,1\}$-SRMT

Definition 4.1 (Multiplicity-free)**.** A graded G-algebra A is *multiplicity-free* if $c_\lambda(A, d) \leq 1$ for all $\lambda \in \Lambda^+$ and $d \geq 0$.

Theorem 4.2 (Spherical $\Rightarrow$ multiplicity-free $\Rightarrow$ $\{0,1\}$-SRMT)**.** *Let $\mathcal{O} \subset W$ be a G-orbit closure.*

- *(a) If $\mathcal{O}$ is G-spherical, then $\mathbf{C}[\mathcal{O}]$ is multiplicity-free as a G-module* (Brion [4])*.*
- *(b) If $\mathbf{C}[\mathcal{O}]$ is multiplicity-free, then $\mathrm{sr}_{\mathcal{O}}(\lambda, \mu; \nu) \in \{0,1\}$ for all (λ, μ, ν).*
- *(c) In the multiplicity-free case, $\mathrm{sr}_{\mathcal{O}}(\lambda, \mu; \nu) = 1$ if and only if $\nu \in \mathcal{X}(\mathcal{O})$ and the product map m_ν is nonzero.*

Proof. (a) Brion's theorem [4]. (b) $\mathrm{sr}_{\mathcal{O}}(\lambda, \mu; \nu) \leq \min(N^\nu_{\lambda\mu} c_\lambda c_\mu, c_\nu) \leq \min(N^\nu_{\lambda\mu}, 1) \in \{0,1\}$. (c) Under multiplicity-freeness the λ- and μ-isotypic components each have unique generators e_λ, e_μ; the SRMT equals 1 iff $m_\nu(e_\lambda \otimes e_\mu) \neq 0$. $\square$

Remark 4.3 (Why Brion's theorem does not apply to the H-action on $\mathcal{O}_{\det_m}$). Theorem 4.2(a) requires $\mathcal{O}$ to be spherical for the same group whose module structure one studies. Here we study $\mathbf{C}[\mathcal{O}_{\det_m}]$ as an H-module. Brion's theorem would require $\mathcal{O}_{\det_m}$ to be spherical for H. This fails for all $m \geq 2$:

$$\dim \mathcal{O}_{\det_m} = m^4 - (2m^2 - 1) = (m^2 - 1)^2, \qquad \dim B_H = m(m+1).$$

For $m = 2$: $\dim \mathcal{O}_{\det_m} = 9 > 6 = \dim B_H$. For $m \geq 3$: $(m^2 - 1)^2 \gg m(m+1)$. In all cases $\dim \mathcal{O}_{\det_m} > \dim B_H$, so no Borel of H has an open orbit in $\mathcal{O}_{\det_m}$.

Full multiplicity-freeness of $\mathbf{C}[\mathcal{O}_{\det_m}]$ as an H-module is Conjecture 4.6 — it is currently an open problem.

Theorem 4.4 (Ray-Multiplicity-1 for the determinant)**.** *For every integer $k \geq 0$, $c_{k\lambda_{\det}}(\mathbf{C}[\mathcal{O}_{\det_m}], k) = 1$. Equivalently, the $k\lambda_{\det}$-isotypic subspace of $(\mathbf{C}[\mathcal{O}_{\det_m}])_k$ is one-dimensional, spanned by $\widehat{\det}_m^{\,k}$.*

Proof. **Step 1 (Upper bound).** We show that $\dim(A_k)_{k\lambda_{\det}} \leq 1$, where $(A_k)_{k\lambda_{\det}}$ denotes the H-weight-$k\lambda_{\det}$ eigenspace in $A_k = \mathbf{C}[\mathcal{O}_{\det_m}]_k$.

By the dual Cauchy decomposition,

$$W^* = \bigoplus_{\lambda \vdash m} S_\lambda(U) \otimes S_\lambda(V) \quad \text{as } H\text{-modules},$$

where $U = V = \mathbf{C}^m$. The component of H-weight $\lambda_{\det} = (\omega_m; \omega_m)$ in W^* is $\wedge^m U \otimes \wedge^m V$, which is *one-dimensional*, spanned by $\widehat{\det}_m$. All other H-isotypic components of W^* have highest weights $(\lambda; \lambda)$ with $\lambda \neq (1^m)$.

The degree-k part of the polynomial ring is $\mathbf{C}[W]_k = \mathrm{Sym}^k(W^*)$. Any element of $\mathrm{Sym}^k(W^*)$ of H-weight $k\lambda_{\det}$ must be a degree-k monomial supported on the $\lambda_{\det}$-weight space of W^*; since that weight space is one-dimensional (spanned by $\widehat{\det}_m$), the unique such monomial is $\widehat{\det}_m^{\,k}$. Hence $\dim(\mathbf{C}[W]_k)_{k\lambda_{\det}} = 1$.

The orbit closure $\mathcal{O}_{\det_m}$ is an affine subvariety of W, so restriction induces an H-equivariant surjection

$$\rho_k : \mathbf{C}[W]_k \twoheadrightarrow A_k.$$

This surjection maps the $k\lambda_{\det}$-eigenspace of $\mathbf{C}[W]_k$ surjectively onto the $k\lambda_{\det}$-eigenspace of A_k. Since the former is at most one-dimensional, so is the latter:

$$c_{k\lambda_{\det}}(\mathbf{C}[\mathcal{O}_{\det_m}], k) = \dim(A_k)_{k\lambda_{\det}} \leq 1.$$

Note: the one-dimensionality of the representation $L^H(k\lambda_{\det})$ alone does not imply this multiplicity bound; the argument uses the one-dimensionality of the $k\lambda_{\det}$-weight space in $\mathbf{C}[W]_k$, which follows from the Cauchy decomposition of W^*.

Step 2 (Non-vanishing). By Remark 2.8, $\widehat{\det}_m \in A_1$ is the unique (up to scalar) H-eigenvector of weight $\lambda_{\det}$ in A_1, and $\widehat{\det}_m^k \in A_k$ is its k-th power, of H-weight $k\lambda_{\det}$. Evaluating at the identity matrix: since $\det_m(I_m) = 1 \neq 0$, the point I_m lies in $\mathcal{O}_{\det_m}$, so $\mathrm{ev}_{I_m}(\widehat{\det}_m^k) = (\det_m(I_m))^k = 1 \neq 0$. Hence $\widehat{\det}_m^k \neq 0$ in A_k, giving $c_{k\lambda_{\det}}(A, k) \geq 1$.

Conclusion. $c_{k\lambda_{\det}}(\mathbf{C}[\mathcal{O}_{\det_m}], k) = 1$ for all $k \geq 0$. □

Remark 4.5. Theorem 4.4 is strictly weaker than full multiplicity-freeness: it asserts $c_{k\lambda_{\det}} = 1$ only for weights along the ray $\{k\lambda_{\det}\}_{k\geq 0}$. This is precisely the information required for Bilinear Rank-1 Rigidity (Theorem 5.4).

Conjecture 4.6 (Full multiplicity-freeness of the determinant orbit closure)**.** *The coordinate ring* $\mathbf{C}[\mathcal{O}_{\det_m}]$ *is multiplicity-free as an* $H = \mathrm{GL}_m \times \mathrm{GL}_m$*-module:* $c_\mu(\mathbf{C}[\mathcal{O}_{\det_m}], d) \leq 1$ *for all* $\mu \in \Lambda^+(H)$ *and* $d \geq 0$.

Status: open. Not required for any result in this paper.

5. Multiplicativity and SRMT Rigidity of the Determinant

Definition 5.1 (Multiplicative polynomial)**.** A polynomial $f \in W$ is (H, χ)-multiplicative for a subgroup $H \leq G$ and character $\chi : H \to \mathbf{C}^*$ if $f(h \cdot x) = \chi(h)f(x)$ for all $h \in H$ and $x \in \mathbf{C}^{m^2}$.

The fundamental example: $\det_m(A^T XB) = \det(A)\det(B)\det_m(X)$, so $\det_m$ is (H, χ)-multiplicative with $\chi(A, B) = \det(A)\det(B)$.

Lemma 5.2 (Multiplicativity Rigidity)**.** *Let* $f \in W$ *be* (H, χ)*-multiplicative with* H*-weight* $\lambda_{\det} = \chi$ *a one-dimensional character of* H*. Assume* $c_{\lambda_{\det}}(\mathbf{C}[\mathcal{O}_f], 1) = 1$*. Then* $\mathrm{sr}^H_{\mathcal{O}_f}(k\lambda_{\det}, k\lambda_{\det}; 2k\lambda_{\det}) = 1$ *for all* $k \geq 1$.

Proof. Case $k = 1$. Since $c_{\lambda_{\det}}(\mathbf{C}[\mathcal{O}_f], 1) = 1$, the $\lambda_{\det}$-isotypic component of $(\mathbf{C}[\mathcal{O}_f])_1$ is $\mathbf{C} \cdot \hat{f}$. The product $\hat{f} \cdot \hat{f} \in A_2$ is an H-eigenvector of weight $2\lambda_{\det}$. Since $\lambda_{\det}$ is a one-dimensional character, $L^H(2\lambda_{\det})$ is one-dimensional, so $c_{2\lambda_{\det}}(A, 2) \leq 1$ (by the same Cauchy argument as in Theorem 4.4, Step 1, applied to $\mathcal{O}_f$). Evaluating at the identity: $\hat{f}^2 \neq 0$, so $c_{2\lambda_{\det}}(A, 2) = 1$. Hence $m_{2\lambda_{\det}}$ is a linear map between one-dimensional spaces sending $\hat{f} \otimes \hat{f} \mapsto \hat{f}^2 \neq 0$: it is an isomorphism of rank 1.

Case $k \geq 2$. The same argument applies to $\hat{f}^k$ and $\hat{f}^{2k}$: both are nonzero H-eigenvectors spanning their respective one-dimensional isotypic components (Theorem 4.4 applied to $\mathcal{O}_f$), and the product map has rank 1. □

Theorem 5.3 (Rank-1 Rigidity along the diagonal ray)**.** $c_{\lambda_{\det}}(\mathcal{O}_{\det_m}, 1) = 1$ *and* $\mathrm{sr}^H_{\mathcal{O}_{\det_m}}(k\lambda_{\det}, k\lambda_{\det}; 2k\lambda_{\det}) = 1$ *for all* $k \geq 1$.

Proof. By Theorem 4.4, $c_{\lambda_{\det}}(\mathbf{C}[\mathcal{O}_{\det_m}], 1) = 1$. Since $\det_m$ is $(H, \det(A)\det(B))$-multiplicative and $\lambda_{\det}$ is a one-dimensional character, Lemma 5.2 applies. □

Theorem 5.4 (Bilinear Rank-1 Rigidity)**.** *For all integers $a, b \geq 0$ with $a + b \geq 1$,*

$$\mathrm{sr}^H_{\mathcal{O}_{\det_m}}(a\lambda_{\det}, b\lambda_{\det}; (a+b)\lambda_{\det}) = 1.$$

Equivalently, the isotypic multiplication map

$$m_{(a+b)\lambda_{\det}} : (\mathbf{C}[\mathcal{O}_{\det_m}]_a)_{a\lambda_{\det}} \otimes (\mathbf{C}[\mathcal{O}_{\det_m}]_b)_{b\lambda_{\det}} \xrightarrow{\sim} (\mathbf{C}[\mathcal{O}_{\det_m}]_{a+b})_{(a+b)\lambda_{\det}}$$

is an isomorphism of one-dimensional $\mathbf{C}$-vector spaces.

Proof. **Step 1 (Multiplicities along the ray).** By Theorem 4.4: $c_{a\lambda_{\det}}(\mathbf{C}[\mathcal{O}_{\det_m}], a) = c_{b\lambda_{\det}}(\mathbf{C}[\mathcal{O}_{\det_m}], b) = c_{(a+b)\lambda_{\det}}(\mathbf{C}[\mathcal{O}_{\det_m}], a+b) = 1$. For each $k \geq 0$, the $k\lambda_{\det}$-isotypic component of A_k is $\mathbf{C} \cdot \widehat{\det}_m^k$. (When $k = 0$, $(\mathbf{C}[\mathcal{O}_{\det_m}]_0)_0 = \mathbf{C} \cdot 1$, the unit of the algebra, and $\widehat{\det}_m^0 = 1$; the case $a = 0$ or $b = 0$ is thus trivial: the multiplication map is the identity on $\mathbf{C} \cdot \widehat{\det}_m^b$ or $\mathbf{C} \cdot \widehat{\det}_m^a$ respectively, with rank 1.)

Step 2 (LR coefficient). Since $\lambda_{\det} = (\omega_m; \omega_m)$ is a one-dimensional character, $L^H(a\lambda_{\det}) \otimes L^H(b\lambda_{\det}) = L^H((a+b)\lambda_{\det})$, so $N^{(a+b)\lambda_{\det}}_{a\lambda_{\det}, b\lambda_{\det}} = 1$ (Remark 2.9).

Step 3 (The map is an isomorphism). The maximum possible SRMT value at $(a\lambda_{\det}, b\lambda_{\det}; (a+b)\lambda_{\det})$ is $\min(1 \cdot 1 \cdot 1, 1) = 1$. The map $m_{(a+b)\lambda_{\det}}$ sends $\widehat{\det}_m^a \otimes \widehat{\det}_m^b \mapsto \widehat{\det}_m^{a+b}$, which is nonzero in A_{a+b} (Step 2 of Theorem 4.4). Hence the rank is 1. □

Remark 5.5 (Source of Rank-1 Rigidity). The key engine of Theorem 5.4 is the one-dimensionality of the $k\lambda_{\det}$-weight space in $\mathbf{C}[W]_k$, which forces $c_{k\lambda_{\det}}(\mathbf{C}[\mathcal{O}_{\det_m}], k) \leq 1$ by surjectivity of restriction; non-vanishing of $\widehat{\det}_m^k$ then establishes equality. This forces $m_{(a+b)\lambda_{\det}}$ to be a map between one-dimensional spaces, so its rank is at most 1; non-vanishing of $\widehat{\det}_m^{a+b}$ promotes this to exactly 1. Neither Brion's theorem nor any sphericity hypothesis is invoked.

Contrast with the permanent: $L^H(k\mu_{\mathrm{perm}}) = \mathrm{Sym}_{kn}(\mathbf{C}^n) \otimes \mathrm{Sym}_{kn}(\mathbf{C}^n)$ has dimension $\binom{kn+n-1}{n}^2 \gg 1$, so the dimensional forcing argument is unavailable. This is a constraint on the present proof method, not a proof that $\mathrm{sr}_{\mathcal{O}_{\mathrm{perm}_n}}$ at the analogous weights cannot equal 1 by some other argument.

Corollary 5.6 (Polynomial sub-algebra along the $\lambda_{\det}$-ray)**.** *The H-isotypic sub-algebra $R_{\lambda_{\det}}(\mathcal{O}_{\det_m}) := \bigoplus_{k \geq 0} (\mathbf{C}[\mathcal{O}_{\det_m}]_k)_{k\lambda_{\det}}$ is isomorphic, as a graded $\mathbf{C}$-algebra, to the polynomial ring $\mathbf{C}[t]$ with $\deg t = 1$.*

Proof. By Theorem 4.4, $(\mathbf{C}[\mathcal{O}_{\det_m}]_k)_{k\lambda_{\det}} = \mathbf{C} \cdot \widehat{\det}_m^k$ for every $k \geq 0$. Theorem 5.4 shows the product map is an isomorphism of one-dimensional graded pieces. Hence $R_{\lambda_{\det}}(\mathcal{O}_{\det_m})$ is freely generated by $\widehat{\det}_m$ with no polynomial relations: $R_{\lambda_{\det}}(\mathcal{O}_{\det_m}) \cong \mathbf{C}[t]$ via $t \mapsto \widehat{\det}_m$. □

Symmetries of the permanent and the dimension obstruction.

Theorem 5.7 (Symmetries of perm_n and the dimension obstruction). *(i)* (Torus acts by character for all $n \geq 2$.) *The diagonal torus $T_H \subset H$ acts on perm_n by the character $\chi_T(t) = (\prod_i t_i)(\prod_j s_j)$.*

(ii) (Stabiliser and dimension for $n \geq 3$.) *For $n \geq 3$, the identity component of $\mathrm{Stab}_{\mathrm{GL}_{n^2}}(\mathrm{perm}_n)$ is the torus $T \cong (\mathbf{C}^*)^{2n-1}$ of row/column rescalings modulo the global scalar, of dimension $2n-1$. Hence*

$$\dim(G \cdot \mathrm{perm}_n) = n^4 - (2n-1) \qquad (n \geq 3).$$

For $n = 2$, see Remark 5.8; the stabiliser is strictly larger and the dimension formula above does not apply.

(iii) (Dimension obstruction to Rank-1 Rigidity.) *For all $n \geq 2$, $\dim L^H(\mu_{\mathrm{perm}}) = \binom{2n-1}{n}^2 \gg 1$. Since $\dim L^H(\mu_{\mathrm{perm}}) > 1$, no analogue of Bilinear Rank-1 Rigidity can be established for $\mathbf{C}[\mathcal{O}_{\mathrm{perm}_n}]$ via the one-dimensionality argument of Theorem 5.4.*

Proof. (i) Each monomial $\prod_i x_{i,\sigma(i)}$ in perm_n scales by $\prod_i t_i \cdot \prod_j s_j$ uniformly over all $\sigma \in S_n$. (ii) For $n \geq 3$: the identity component of the stabiliser of perm_n in GL_{n^2} is the torus of row/column rescalings, of dimension $2n-1$, as established in the proof of Proposition 8.1. For $n = 2$: see Remark 5.8. (iii) Follows from Proposition 2.6(iii). □

Remark 5.8 (The case $n = 2$: perm_2 as a quadratic form). The polynomial $\mathrm{perm}_2(X) = x_{11}x_{22} + x_{12}x_{21}$ is a non-degenerate symmetric bilinear form in four variables $(x_{11}, x_{12}, x_{21}, x_{22}) \in \mathbf{C}^4$. Its stabiliser in $G = \mathrm{GL}_4(\mathbf{C})$ is the complex orthogonal group $\mathrm{O}(4, \mathbf{C})$, which has dimension $\binom{4}{2} = 6$. Therefore:

$$\dim(G \cdot \mathrm{perm}_2) = 16 - 6 = 10 = \dim B_4,$$

where $B_4 \subset \mathrm{GL}_4$ is a Borel subgroup (dimension $\frac{4\cdot 5}{2} = 10$).

We now show that $G \cdot \mathrm{perm}_2$ is spherical. The orbit $G \cdot \mathrm{perm}_2 \cong \mathrm{GL}_4(\mathbf{C})/\mathrm{O}(4, \mathbf{C})$ is the variety of non-degenerate symmetric bilinear forms on $\mathbf{C}^4$. The Borel subgroup $B_4 \subset \mathrm{GL}_4(\mathbf{C})$ of upper-triangular matrices acts on symmetric forms by $(b, q) \mapsto bqb^T$. Over $\mathbf{C}$ (algebraically closed, characteristic $\neq 2$), the LDL^T factorisation (Cholesky over a field) shows that every non-degenerate symmetric bilinear form admits a decomposition $q = b \cdot q_0 \cdot b^T$ with $b \in B_4$ upper-triangular and q_0 diagonal; since all non-degenerate diagonal forms over $\mathbf{C}$ are GL_4-equivalent to the identity (as $\mathbf{C}$ is algebraically closed), one obtains

that B_4 acts transitively on $G \cdot \mathrm{perm}_2$. Hence B_4 has an open dense orbit, and $G \cdot \mathrm{perm}_2$ is spherical. See Brion [5] for the classification of spherical homogeneous spaces, where $\mathrm{GL}_n/\mathrm{O}_n$ appears as a standard example of a symmetric space that is spherical.

Consequently:

(a) The non-sphericity claim of Proposition 8.1 does not apply to $n = 2$.
(b) The dimension formula $\dim(G \cdot \mathrm{perm}_n) = n^4 - (2n - 1)$ is incorrect for $n = 2$ (it would give 13, not 10).
(c) For the SRMT separation programme, $n = 2$ requires separate treatment. The H-module distinction between $\det_2$ and perm_2 persists (they lie in different Schur components of the Cauchy decomposition, with $\lambda_{\det} = (\omega_2; \omega_2)$ one-dimensional and $\mu_{\mathrm{perm}} = (2\omega_1; 2\omega_1)$ of dimension $\binom{3}{2}^2 = 9$), so the H-module-level structural argument of §9 applies to $n = 2$ via a different route, independently of sphericity.

Remark 5.9 (Failure of Rank-1 Rigidity for perm_n). For $\mu_{\mathrm{perm}} = (m\omega_1; m\omega_1)$ with $m \geq 2$, $\dim L^H(\mu_{\mathrm{perm}}) = \binom{2m-1}{m}^2 > 1$ and $N^{2\mu_{\mathrm{perm}}}_{\mu_{\mathrm{perm}},\mu_{\mathrm{perm}}} = 1$ (Pieri; Remark 2.9). Consequently:

(i) The SRMT upper bound at $(a\mu_{\mathrm{perm}}, b\mu_{\mathrm{perm}}; (a + b)\mu_{\mathrm{perm}})$ is $\min(c_{a\mu_{\mathrm{perm}}} \cdot c_{b\mu_{\mathrm{perm}}}, c_{(a+b)\mu_{\mathrm{perm}}})$ (using $N = 1$). If $c_{\mu_{\mathrm{perm}}} \geq 2$ (Open Problem (S1)), this upper bound is ≥ 2, meaning SRMT values above 1 are not ruled out.
(ii) The multiplicity $c_{k\mu_{\mathrm{perm}}}(\mathbf{C}[\mathcal{O}_{\mathrm{perm}_n}], k)$ is not constrained to ≤ 1 by any character identity.
(iii) The analogue of Corollary 5.6 does not follow for $\mathbf{C}[\mathcal{O}_{\mathrm{perm}_n}]$ in general.

Statements (i)–(iii) establish that the *mechanism* of Bilinear Rank-1 Rigidity (one-dimensionality forcing) is unavailable for the permanent. They do not by themselves prove that the SRMT of $\mathbf{C}[\mathcal{O}_{\mathrm{perm}_n}]$ at $(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; 2\mu_{\mathrm{perm}})$ actually exceeds 1; that requires (S1) and (S2).

6. SRMT Defects and the Structure of Syzygies

Definition 6.1 (Isotypic syzygy and cosyzygy at the multiplicity level)**.** For a graded G-algebra A, fixed degrees $d_1, d_2 \geq 1$, and weights $\lambda, \mu, \nu \in \Lambda^+$, let $\widetilde{m}_\nu$ denote the multiplicity-space map of Definition 2.3:

$$\widetilde{m}_\nu : \mathbf{C}^{N^\nu_{\lambda\mu} c_\lambda(A,d_1) c_\mu(A,d_2)} \to \mathbf{C}^{c_\nu(A,d_1+d_2)}.$$

The *isotypic syzygy space* and *isotypic cosyzygy space* are, respectively:

$$\widetilde{\mathrm{Syz}}^{d_1,d_2}(A)_{\lambda,\mu;\nu} := \ker\big(\widetilde{m}_\nu\big),$$
$$\widetilde{\mathrm{CoSyz}}^{d_1,d_2}(A)_{\lambda,\mu;\nu} := \mathrm{coker}\big(\widetilde{m}_\nu\big).$$

These are vector spaces (not G-modules) of dimensions $N^{\nu}_{\lambda\mu}c_\lambda c_\mu - \mathrm{sr}^{d_1,d_2}_A(\lambda,\mu;\nu)$ and $c_\nu - \mathrm{sr}^{d_1,d_2}_A(\lambda,\mu;\nu)$, respectively.

If one wishes to work with the actual G-module kernels and cokernels, let

$$\mathrm{Syz}^{d_1,d_2}(A)_{\lambda,\mu;\nu} := \ker\big(m_\nu : (A_{d_1})_\lambda \otimes (A_{d_2})_\mu \to (A_{d_1+d_2})_\nu\big),$$
$$\mathrm{CoSyz}^{d_1,d_2}(A)_{\lambda,\mu;\nu} := \mathrm{coker}\big(m_\nu : (A_{d_1})_\lambda \otimes (A_{d_2})_\mu \to (A_{d_1+d_2})_\nu\big).$$

These satisfy $\mathrm{Syz}^{d_1,d_2}(A)_{\lambda,\mu;\nu} \cong \widetilde{\mathrm{Syz}}^{d_1,d_2}(A)_{\lambda,\mu;\nu} \otimes_{\mathbf{C}} L(\nu)$ as G-modules.

Theorem 6.2 (SRMT–Syzygy exact sequence)**.** *For any graded G-algebra A and $\lambda,\mu,\nu \in \Lambda^+$:*

$$0 \to \widetilde{\mathrm{Syz}}^{d_1,d_2}(A)_{\lambda,\mu;\nu} \to \mathbf{C}^{N^{\nu}_{\lambda\mu}c_\lambda(A,d_1)c_\mu(A,d_2)} \xrightarrow{\widetilde{m}_\nu} \mathbf{C}^{c_\nu(A,d_1+d_2)} \to \widetilde{\mathrm{CoSyz}}^{d_1,d_2}(A)_{\lambda,\mu;\nu} \to 0,$$

with (all dimensions being multiplicity-space dimensions):

$$\begin{aligned}\mathrm{sr}^{d_1,d_2}_A(\lambda,\mu;\nu) &= N^{\nu}_{\lambda\mu}c_\lambda(A,d_1)c_\mu(A,d_2) - \dim \widetilde{\mathrm{Syz}}^{d_1,d_2}(A)_{\lambda,\mu;\nu}\\ &= c_\nu(A,d_1+d_2) - \dim \widetilde{\mathrm{CoSyz}}^{d_1,d_2}(A)_{\lambda,\mu;\nu}.\end{aligned}$$

At the G-module level, the analogous exact sequence is:

$$0 \to \mathrm{Syz}^{d_1,d_2}(A)_{\lambda,\mu;\nu} \to (A_{d_1})_\lambda \otimes (A_{d_2})_\mu \xrightarrow{m_\nu} (A_{d_1+d_2})_\nu \to \mathrm{CoSyz}^{d_1,d_2}(A)_{\lambda,\mu;\nu} \to 0,$$

with G-module rank-nullity:

$$\mathrm{rank}(m_\nu|_{(A_{d_1})_\lambda\otimes(A_{d_2})_\mu}) = c_\lambda(A,d_1)\dim L(\lambda)\cdot c_\mu(A,d_2)\dim L(\mu) - \dim \mathrm{Syz}^{d_1,d_2}(A)_{\lambda,\mu;\nu},$$

where $\mathrm{sr}^{d_1,d_2}_A(\lambda,\mu;\nu) = \mathrm{rank}(m_\nu|_{\dots})/\dim L(\nu)$.

Proof. Standard rank-nullity applied to $\widetilde{m}_\nu$. □

Corollary 6.3 (SRMT under degree-1 generation)**.** *Let A be a connected graded G-algebra generated in degree 1. Then the global multiplication map $m_{1,1} : A_1 \otimes A_1 \twoheadrightarrow A_2$ is surjective. However, generation in degree 1 does not imply individual surjectivity $\widetilde{\mathrm{CoSyz}}^{1,1}(A)_{\lambda_0,\mu_0;\nu} = 0$ at a fixed pair (λ_0,μ_0). The individual SRMT defect $\delta^{1,1}_A(\lambda,\mu;\nu)$ can be positive even when A is generated in degree 1.*

7. SRMT Monotonicity

Theorem 7.1 (SRMT Monotonicity at fixed degree pairs)**.** *Let $\mathcal{O}_1 \subseteq \mathcal{O}_2$ be G-orbit closures in W, and let $d_1,d_2 \geq 0$ be fixed integers. Then*

$$\mathrm{sr}^{d_1,d_2}_{\mathcal{O}_1}(\lambda,\mu;\nu) \leq \mathrm{sr}^{d_1,d_2}_{\mathcal{O}_2}(\lambda,\mu;\nu) \quad \textit{for all } \lambda,\mu,\nu \in \Lambda^+.$$

Proof. See Appendix C. □

Remark 7.2. Monotonicity holds for each fixed degree pair (d_1,d_2). Comparison at respective minimal degrees (which can differ between the two algebras) falls outside this theorem.

Corollary 7.3. *If $\mathrm{sr}^{d_1,d_2}_{\mathcal{O}_1}(\lambda_0,\mu_0;\nu_0) > \mathrm{sr}^{d_1,d_2}_{\mathcal{O}_2}(\lambda_0,\mu_0;\nu_0)$ for a single triple $(\lambda_0,\mu_0;\nu_0)$ and fixed (d_1,d_2), then $\mathcal{O}_1 \not\subseteq \mathcal{O}_2$.*

8. The Permanent's SRMT

Proposition 8.1 (Non-sphericity of $\mathcal{O}_{\mathrm{perm}_n}$ for $n \geq 3$)**.** *For all $n \geq 3$, the orbit $G \cdot \mathrm{perm}_n$ is not spherical for $G = \mathrm{GL}_{n^2}$. Consequently, Brion's criterion for multiplicity-freeness of $\mathbf{C}[\mathcal{O}_{\mathrm{perm}_n}]$ as a G-module is not satisfied for $n \geq 3$.*

The case $n = 2$ is exceptional: $G \cdot \mathrm{perm}_2$ is spherical (see Remark 5.8).

Proof. For $n \geq 3$: the identity component of the stabiliser of perm_n in $G = \mathrm{GL}_{n^2}$ is the torus $(\mathbf{C}^*)^{2n-1}$ of independent row and column rescalings modulo the global scalar. Explicitly, for $i = 1, \ldots, n$ and $j = 1, \ldots, n$, the rescaling $(x_{ij} \mapsto r_i c_j x_{ij})$ preserves perm_n for any $r_i, c_j \in \mathbf{C}^*$, since each monomial $\prod_i x_{i,\sigma(i)}$ acquires the same factor $(\prod_i r_i)(\prod_j c_j)$. This gives a torus of dimension $2n$; modding out the global scalar ($r_i = c_j = t$) yields dimension $2n - 1$. For $n \geq 3$, Grochow [2] (Theorem 3.1 and the discussion of stabilisers of the permanent in Chapter 4) confirms that no other continuous symmetry exists, so the identity component of $\mathrm{Stab}_G(\mathrm{perm}_n)$ is exactly this torus. Hence $\dim(G \cdot \mathrm{perm}_n) = n^4 - (2n - 1)$.

A Borel subgroup $B_{n^2} \subset \mathrm{GL}_{n^2}$ has dimension $\frac{n^2(n^2+1)}{2}$. We compare:

n	$\dim(G \cdot \mathrm{perm}_n)$	$\dim B_{n^2}$	Spherical?
3	$81 - 5 = 76$	$\frac{9 \cdot 10}{2} = 45$	No ($76 > 45$)
4	$256 - 7 = 249$	$\frac{16 \cdot 17}{2} = 136$	No ($249 > 136$)

For general $n \geq 3$, the inequality $n^4 - (2n-1) > \frac{n^2(n^2+1)}{2}$ simplifies to $n^4 - n^2 - 4n + 2 > 0$. For $n \geq 3$: $n^4 - n^2 - 4n + 2 \geq 81 - 9 - 12 + 2 = 62 > 0$. Since $\dim(G \cdot \mathrm{perm}_n) > \dim B_{n^2}$ for all $n \geq 3$, no Borel of GL_{n^2} has an open orbit, and the orbit is not spherical. □

Remark 8.2 (Non-sphericity vs. non-multiplicity-freeness)*.* Proposition 8.1 establishes that the sufficient condition for multiplicity-freeness given by Brion's theorem is not fulfilled for $n \geq 3$. The converse of Brion's theorem is false in general. Whether $\mathbf{C}[\mathcal{O}_{\mathrm{perm}_n}]$ actually has some $c_\lambda \geq 2$ is a separate open question (Open Problem (S1)).

Remark 8.3 (SRMT flexibility for perm_n, conditional on (S1))*.* Suppose $c_{\mu_{\mathrm{perm}}}(\mathbf{C}[\mathcal{O}_{\mathrm{perm}_n}], d_0) \geq 2$ for some degree d_0 (Open Problem (S1)). Then the maximum possible SRMT at $(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; 2\mu_{\mathrm{perm}})$ at degree pair (d_0, d_0) is $\min(c^2_{\mu_{\mathrm{perm}}}, c_{2\mu_{\mathrm{perm}}}) \geq 2$ (using $N^{2\mu_{\mathrm{perm}}}_{\mu_{\mathrm{perm}},\mu_{\mathrm{perm}}} = 1$). Since no character identity constrains the multiplication maps (Theorem 5.7), there is no algebraic mechanism forcing the SRMT strictly below its theoretical upper bound. This does not prove the SRMT is ≥ 2; it shows there is no a priori obstruction to it being ≥ 2. Verifying this computationally is Open Problem (S2).

Theorem 8.4 (Structural asymmetry between determinant and permanent)**.** *Let $m \geq n \geq 2$.*

(i) (Unconditional.) *For every $a, b \geq 0$ with $a+b \geq 1$: $\mathrm{sr}^H_{\mathcal{O}_{\det_m}}(a\lambda_{\det}, b\lambda_{\det}; (a+b)\lambda_{\det}) = 1$. The proof of Bilinear Rank-1 Rigidity relies on the one-dimensionality of $L^H(k\lambda_{\det})$ for all $k \geq 0$. Since $\dim L^H(\mu_{\mathrm{perm}}) = \binom{2n-1}{n}^2 \gg 1$, the same argument cannot establish an analogous result for $\mathbf{C}[\mathcal{O}_{\mathrm{perm}_n}]$ at the weights $(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; 2\mu_{\mathrm{perm}})$. This is an asymmetry in the available proof technique; it does not exclude the possibility that $\mathrm{sr}^H_{\mathcal{O}_{\mathrm{perm}_n}}(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; 2\mu_{\mathrm{perm}}) = 1$ by a different argument.*

(ii) (Unconditional, for $n \geq 3$.) *The orbit $G \cdot \mathrm{perm}_n$ is not spherical for $G = \mathrm{GL}_{n^2}$; Brion's multiplicity-free criterion fails for $\mathbf{C}[\mathcal{O}_{\mathrm{perm}_n}]$. For $n = 2$, the orbit is spherical (Remark 5.8).*

(iii) (Conditional on (S1)+(S2).) *If $c_{\mu_{\mathrm{perm}}}(\mathbf{C}[\mathcal{O}_{\mathrm{perm}_n}], d_0) \geq 2$ and the multiplication matrix has rank ≥ 2, then $\mathrm{sr}^{d_0,d_0}_{\mathcal{O}_{\mathrm{perm}_n}}(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; 2\mu_{\mathrm{perm}}) \geq 2$.*

(iv) (Unconditional.) *SRMT values are monotone under orbit-closure inclusion at any fixed degree pair.*

9. The $n = 3$ Specialisation and the Computational Programme

For $n = m = 3$ all ingredients are finite-dimensional: $G = \mathrm{GL}_9$, $W = \mathrm{Sym}_3(\mathbf{C}^{9*})$ ($\dim W = \binom{11}{3} = 165$). The Cauchy decomposition gives three H-components:

Component	H-weight	Contains	dim
(1^3): $\wedge^3 U^* \otimes \wedge^3 V^*$	$\lambda_{\det} = (\omega_3; \omega_3)$	$\det_3$	1
$(2,1)$: $S_{(2,1)}(U^*) \otimes S_{(2,1)}(V^*)$	$((2,1,0); (2,1,0))$	—	64
(3): $\mathrm{Sym}_3 U^* \otimes \mathrm{Sym}_3 V^*$	$\mu_{\mathrm{perm}} = (3\omega_1; 3\omega_1)$	perm_3	100

9.1. **The structural gap at $n = 3$.** The proved unconditional results specialise as follows.

- $\mathrm{sr}^H_{\mathcal{O}_{\det_m}}(a\lambda_{\det}, b\lambda_{\det}; (a+b)\lambda_{\det}) = 1$ for all $a, b \geq 0$, $a + b \geq 1$ (Theorem 5.4).
- $G \cdot \mathrm{perm}_3$ is not spherical for $G = \mathrm{GL}_9$ (Proposition 8.1).
- No character identity constrains the SRMT of $\mathbf{C}[\mathcal{O}_{\mathrm{perm}_n}]$ (Theorem 5.7).

9.2. **Open problems: the computational separation programme.** The SRMT framework leads to the following computational programme. If all three hypotheses below are verified (feasible for $n = 3$ using Macaulay2; see Appendix B), one obtains a *finite non-inclusion* $\mathcal{O}_{\mathrm{perm}_3} \not\subseteq \mathcal{O}_{\det_3}$.

(S1) Non-trivial multiplicity for the permanent. Compute the H-isotypic decomposition of $\mathbf{C}[\mathcal{O}_{\mathrm{perm}_3}]_d$ for $d = 1, 2, 3, \ldots$ Determine the smallest degree $d_0 \geq 2$ at which $c_{\mu_{\mathrm{perm}}}(\mathbf{C}[\mathcal{O}_{\mathrm{perm}_3}], d_0) > 0$, and verify whether $c_{\mu_{\mathrm{perm}}}(\mathbf{C}[\mathcal{O}_{\mathrm{perm}_3}], d_0) \geq 2$. *Status: open.*

(S2) SRMT value ≥ 2 for the permanent. Assuming (S1) holds with $r := c_{\mu_{\mathrm{perm}}}(\mathbf{C}[\mathcal{O}_{\mathrm{perm}_3}], d_0) \geq 2$, construct the $r \times r$ multiplication matrix and verify $\mathrm{rank} M \geq 2$. *Status: open; depends on (S1).*

(S3) SRMT value ≤ 1 for the determinant at $(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; \nu^*)$. Compute $c_{\nu^*}(\mathbf{C}[\mathcal{O}_{\det_3}])$ for $\nu^* = 2\mu_{\mathrm{perm}} = (6\omega_1; 6\omega_1)$. Since $\mu_{\mathrm{perm}} \neq k\lambda_{\det}$ for any k (Remark 2.9), this does not follow from Theorem 4.4 and requires separate computation. *Status: open.*

(S4) Orbit-level multiplicity–SRMT independence. Do there exist G-orbit closures $\mathcal{O}_1, \mathcal{O}_2 \subset W$ with $c_\lambda(\mathbf{C}[\mathcal{O}_1], d) = c_\lambda(\mathbf{C}[\mathcal{O}_2], d)$ for all (λ, d), yet $\mathrm{sr}^{d_1,d_2}_{\mathcal{O}_1}(\lambda, \mu; \nu) \neq \mathrm{sr}^{d_1,d_2}_{\mathcal{O}_2}(\lambda, \mu; \nu)$ for some tuple? *Status: open; not required for the separation via (S1)–(S3).*

Logical conclusion (corrected). If (S1), (S2), and (S3) all hold, then by SRMT Monotonicity (Corollary 7.3) at degree pair (d_0, d_0):

$$\mathrm{sr}^{d_0,d_0}_{\mathcal{O}_{\mathrm{perm}_3}}(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; \nu^*) \geq 2 > 1 \geq \mathrm{sr}^{d_0,d_0}_{\mathcal{O}_{\det_3}}(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; \nu^*), \tag{1}$$

which gives $\mathcal{O}_{\mathrm{perm}_3} \not\subseteq \mathcal{O}_{\det_3}$.

This is a finite non-inclusion result for the specific values $n = m = 3$. It does *not* imply Valiant's conjecture (VP $\neq$ VNP), which requires the non-inclusion $\mathcal{O}_{\mathrm{perm}_n} \not\subseteq \mathcal{O}_{\det_m}$ for *all* n with m polynomial in n — an asymptotic family of statements. The case $n = m = 3$ is one data point in such a programme, not a proof of the conjecture.

10. Perspectives

10.1. Koszul structure of $\mathbf{C}[\mathcal{O}_{\det_m}]$.

Definition 10.1 (Koszul algebra; Priddy [10])**.** A connected graded $\mathbf{C}$-algebra $A = \bigoplus_{d \geq 0} A_d$, generated in degree 1, is *Koszul* if $\mathrm{Ext}^i_A(\mathbf{C}, \mathbf{C})_j = 0$ for all $i \neq j$.

Remark 10.2 (Generation in degree 1 for $\mathcal{O}_{\det_m}$ — hypothesis K1). The variety $\mathcal{O}_{\det_m}$ is an affine cone in W, so $\mathbf{C}[\mathcal{O}_{\det_m}]$ is a quotient of the polynomial ring $\mathbf{C}[W]$, which is generated in degree 1. For $m = 2$ (classical Plücker), (K1) holds. For $m \geq 3$, (K1) is expected by analogy with secant variety theory, but is not proved. *Status: proved for $m = 2$; expected for $m \geq 3$; decidable for $m = 3$ via Macaulay2 (Script 2, Appendix B).*

Conjecture 10.3 (H-equivariant Koszul property)**.** *Assume the following prerequisites hold for $\mathbf{C}[\mathcal{O}_{\det_m}]$:*

(K1) $\mathbf{C}[\mathcal{O}_{\det_m}]$ *is generated in degree* 1.
(K2) *The ideal $I(\mathcal{O}_{\det_m})$ is generated in degree* 2.

Then $\mathbf{C}[\mathcal{O}_{\det_m}]$ *is* H*-equivariantly Koszul. For* $m = 2$*, (K2) is proved. For* $m \geq 3$*, (K2) is expected to fail; the* N*-Koszul theory of Berger* [3] *provides a natural extension in that case. Status: both (K1) and (K2) for* $m \geq 3$ *are open; the conjecture is conditional on both.*

10.2. **Weight semigroup of** $\mathcal{O}_{\det_m}$**.**

Conjecture 10.4 (Weight semigroup under H)**.** $\mathcal{X}_H(\mathcal{O}_{\det_m}) = \{(\lambda;\lambda) \in \Lambda^+(H) : \ell(\lambda) \leq m,\ \lambda_1 \leq m\}$*, i.e.,* H*-dominant weights of the form* $(\lambda;\lambda)$ *where* λ *fits in an* $m \times m$ *box. Status: open.*

10.3. **Newton–Okounkov bodies.**

Conjecture 10.5 (Newton–Okounkov volume separation)**.** *For all* $n \geq 3$*:* $\operatorname{vol}\Delta(\mathcal{O}_{\mathrm{perm}_n}) > \operatorname{vol}\Delta(\mathcal{O}_{\det_m})$*. Status: open; no evidence beyond dimensional heuristics.*

11. Conclusion

What has been established (unconditionally).

(1) **Module–algebra gap** (Proposition 3.1). Graded G-algebras with identical isotypic multiplicity tables can have distinct SRMT values. BIP does not obstruct SRMT-based separation arguments for abstract graded algebras.
(2) **Ray-Multiplicity-1 and Bilinear Rank-1 Rigidity of the determinant** (Theorems 4.4 and 5.4). $c_{k\lambda_{\det}}(\mathbf{C}[\mathcal{O}_{\det_m}], k) = 1$ for all $k \geq 0$, proved directly from the Cauchy decomposition without any sphericity assumption. The $\lambda_{\det}$-ray isotypic subalgebra is $\cong \mathbf{C}[t]$.
(3) **Dimension obstruction for the permanent, and corrected non-sphericity of its orbit** (Theorem 5.7 and Proposition 8.1). For $n \geq 3$, $G \cdot \mathrm{perm}_n$ is not spherical; for $n = 2$, the orbit is spherical (Remark 5.8). The dimension $\dim L^H(\mu_{\mathrm{perm}}) \gg 1$ prevents any analogue of Bilinear Rank-1 Rigidity via the one-dimensionality argument.
(4) **SRMT–syzygy correspondence** (Theorem 6.2).
(5) **SRMT Monotonicity** (Theorem 7.1 and Appendix C), with an explicit proof verifying surjectivity on isotypic components via G-equivariance and Schur's lemma.

What is conditional or conjectural.

- The finite non-inclusion $\mathcal{O}_{\mathrm{perm}_3} \not\subseteq \mathcal{O}_{\det_3}$ is conditional on (S1)+(S2)+(S3).
- Full multiplicity-freeness of $\mathbf{C}[\mathcal{O}_{\det_m}]$ is Conjecture 4.6 (open).
- Koszul structure is Conjecture 10.3 (conditional on K1+K2, both open for $m \geq 3$).
- Weight semigroup description is Conjecture 10.4 (open).
- Newton–Okounkov volume separation is Conjecture 10.5 (open).
- An asymptotic SRMT-based approach to VP $\neq$ VNP is a long-term research programme, not a result of this paper.

Appendix A. Notation Summary

Symbol	Meaning
$G = \mathrm{GL}_{m^2}$	Full linear group acting on W
$W = \mathrm{Sym}_m(\mathbf{C}^{m^2*})$	Space of degree-m forms in m^2 variables
$H = \mathrm{GL}_m \times \mathrm{GL}_m$	Subgroup via $(A,B)\cdot X = AXB^T$
$\mathcal{O}_{\det_m} = \overline{G\cdot \det_m}$	G-orbit closure of $\det_m$ (Zariski)
$\mathcal{O}_{\mathrm{perm}_n} = \overline{G\cdot \mathrm{perm}_n}$	G-orbit closure of perm_n (Zariski)
Λ^+	Dominant integral weights of G (or H)
$L(\lambda)$	Irreducible G-module of highest weight λ
$c_\lambda(V)$	Multiplicity $\dim \mathrm{Hom}_G(L(\lambda), V)$
$N^\nu_{\lambda\mu}$	Littlewood–Richardson coefficient
$\mathrm{sr}_A^{d_1,d_2}(\lambda,\mu;\nu)$	SRMT value at degree pair (d_1,d_2)
$\delta_A^{d_1,d_2}(\lambda,\mu;\nu)$	SRMT defect
$\mathcal{X}(\mathcal{O})$	Weight semigroup of $\mathcal{O}$
$\lambda_{\det} = (\omega_m;\omega_m)$	H-highest weight of $\det_m$ (one-dimensional)
$\mu_{\mathrm{perm}} = (m\omega_1; m\omega_1)$	H-highest weight of perm_n
$\widehat{\det}_m \in A_1$	Coordinate-ring element of H-weight $\lambda_{\det}$ (distinct from $\det_m \in W$)
$\widetilde{\mathrm{Syz}}^{d_1,d_2}(A)_{\lambda,\mu;\nu}$	Isotypic syzygy kernel (multiplicity-space level)
$\widetilde{\mathrm{CoSyz}}^{d_1,d_2}(A)_{\lambda,\mu;\nu}$	Isotypic cosyzygy cokernel (multiplicity-space level)
$R_{\lambda_{\det}}(\mathcal{O}_{\det_m})$	$\lambda_{\det}$-ray isotypic sub-algebra $\cong \mathbf{C}[t]$

Appendix B. Computational Programme for $n = 3$

The following describes a concrete computational programme to verify hypotheses (S1)–(S3) for $n = m = 3$. **These computations have not been carried out by the author.** Their execution and reporting of numerical outputs would constitute essential evidence for or against the conditional separation result of §9.

Setup. Work over in the polynomial ring $R = [x_{11}, x_{12}, x_{13}, x_{21}, x_{22}, x_{23}, x_{31}, x_{32}, x_{33}]$ with the 3×3 permanent and determinant:

$$\mathrm{perm}_3 = x_{11}x_{22}x_{33} + x_{11}x_{23}x_{32} + x_{12}x_{21}x_{33} + x_{12}x_{23}x_{31} + x_{13}x_{21}x_{32} + x_{13}x_{22}x_{31},$$
$$\det_3 = x_{11}x_{22}x_{33} - x_{11}x_{23}x_{32} - x_{12}x_{21}x_{33} + x_{12}x_{23}x_{31} + x_{13}x_{21}x_{32} - x_{13}x_{22}x_{31}.$$

The $H = \mathrm{GL}_3 \times \mathrm{GL}_3$-weights of these polynomials in the Cauchy decomposition of $W = \mathrm{Sym}_3(\mathbf{C}^{9*})$ are:

$$\det_3 \in S_{(1^3)}(\mathbf{C}^{3*}) \otimes S_{(1^3)}(\mathbf{C}^{3*}), \quad \lambda_{\det} = (\omega_3;\omega_3), \quad \dim = 1;$$
$$\mathrm{perm}_3 \in \mathrm{Sym}_3(\mathbf{C}^{3*}) \otimes \mathrm{Sym}_3(\mathbf{C}^{3*}), \quad \mu_{\mathrm{perm}} = (3\omega_1; 3\omega_1), \quad \dim = 100.$$

Note that both polynomials have the same torus character $(1,1,1;1,1,1)$; the distinction is at the Schur-module level.

Programme for (S1): non-trivial multiplicity for the permanent. Using the `SchurRings` package in Macaulay2 [14], compute the H-isotypic decomposition of $\mathbf{C}[\mathcal{O}_{\mathrm{perm}_3}]_d$ for $d = 1, 2, 3, \ldots$ by computing the coordinate ring of the orbit closure (as the quotient of $\mathbf{C}[W]$ by the defining ideal of $\overline{G \cdot \mathrm{perm}_3}$) and decomposing each graded piece as an H-module. Determine the smallest $d_0 \geq 2$ at which the multiplicity $c_{\mu_{\mathrm{perm}}}(\mathbf{C}[\mathcal{O}_{\mathrm{perm}_3}], d_0) \geq 2$.

Programme for (S2): SRMT value ≥ 2. Assuming (S1) yields $r = c_{\mu_{\mathrm{perm}}}(\mathbf{C}[\mathcal{O}_{\mathrm{perm}_3}], d_0) \geq 2$: extract a basis $\{v_1, \ldots, v_r\}$ for the μ_{perm}-isotypic subspace of $(\mathbf{C}[\mathcal{O}_{\mathrm{perm}_3}])_{d_0}$. Form the $r \times r$ multiplication matrix M with M_{ij} = the coordinate of $v_i \cdot v_j$ in $(\mathbf{C}[\mathcal{O}_{\mathrm{perm}_3}]_{2d_0})_{2\mu_{\mathrm{perm}}}$ (with respect to a fixed basis of that subspace). Compute $\mathrm{rank} M$ and verify $\mathrm{rank} M \geq 2$.

Programme for (S3): SRMT ≤ 1 for the determinant. Compute $c_{\nu^*}(\mathbf{C}[\mathcal{O}_{\det_3}])$ for $\nu^* = 2\mu_{\mathrm{perm}} = (6\omega_1; 6\omega_1)$. Since $\mu_{\mathrm{perm}} \neq k\lambda_{\det}$ for any k (Remark 2.9), this does not follow from Theorem 4.4 and requires a separate isotypic decomposition computation.

Consequence. If (S1), (S2), and (S3) are all verified:

$$\mathrm{sr}^{d_0,d_0}_{\mathcal{O}_{\mathrm{perm}_3}}(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; 2\mu_{\mathrm{perm}}) \geq 2 > 1 \geq \mathrm{sr}^{d_0,d_0}_{\mathcal{O}_{\det_3}}(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; 2\mu_{\mathrm{perm}}),$$

which by Corollary 7.3 gives $\mathcal{O}_{\mathrm{perm}_3} \not\subseteq \mathcal{O}_{\det_3}$. This is a finite non-inclusion for $n = m = 3$ only and does not imply VP $\neq$ VNP.

Appendix C. Proof of SRMT Monotonicity

Theorem C.1 (SRMT Monotonicity at fixed degree pairs)**.** *Let $\mathcal{O}_1 \subseteq \mathcal{O}_2$ be G-orbit closures in W. For any fixed integers $d_1, d_2 \geq 0$:*

$$\mathrm{sr}^{d_1,d_2}_{\mathcal{O}_1}(\lambda, \mu; \nu) \leq \mathrm{sr}^{d_1,d_2}_{\mathcal{O}_2}(\lambda, \mu; \nu) \quad \textit{for all } \lambda, \mu, \nu \in \Lambda^+.$$

Proof. **Step 1 (The surjection).** The Zariski-closed inclusion $\mathcal{O}_1 \subseteq \mathcal{O}_2$ induces a surjection of graded G-algebras $\varphi : \mathbf{C}[\mathcal{O}_2] \twoheadrightarrow \mathbf{C}[\mathcal{O}_1]$, $f \mapsto f|_{\mathcal{O}_1}$, which is G-equivariant and respects the grading.

Step 2 (Surjectivity on isotypic components). Fix $d \geq 0$ and $\lambda \in \Lambda^+$. The degree-d component $\varphi_d : \mathbf{C}[\mathcal{O}_2]_d \twoheadrightarrow \mathbf{C}[\mathcal{O}_1]_d$ is a surjective G-module map between finite-dimensional rational G-modules. Write the isotypic decompositions $\mathbf{C}[\mathcal{O}_2]_d = \bigoplus_\lambda V_\lambda$ and $\mathbf{C}[\mathcal{O}_1]_d = \bigoplus_\lambda U_\lambda$. Since φ_d is G-equivariant, it maps each isotypic component V_λ into U_λ by Schur's lemma. If any block $(\varphi_d)_\lambda : V_\lambda \to U_\lambda$ were not surjective, the cokernel of φ_d would contain a nonzero $L(\lambda)$-isotypic part, contradicting global surjectivity. Hence each $(\varphi_d)_\lambda$ is surjective.

Applying this with $d = d_1$, $d = d_2$, $d = d_1 + d_2$:

$$\alpha := (\varphi_{d_1})_\lambda : (\mathbf{C}[\mathcal{O}_2]_{d_1})_\lambda \twoheadrightarrow (\mathbf{C}[\mathcal{O}_1]_{d_1})_\lambda,$$
$$\beta := (\varphi_{d_2})_\mu : (\mathbf{C}[\mathcal{O}_2]_{d_2})_\mu \twoheadrightarrow (\mathbf{C}[\mathcal{O}_1]_{d_2})_\mu,$$
$$\gamma := (\varphi_{d_1+d_2})_\nu : (\mathbf{C}[\mathcal{O}_2]_{d_1+d_2})_\nu \twoheadrightarrow (\mathbf{C}[\mathcal{O}_1]_{d_1+d_2})_\nu.$$

Step 3 (Commutativity with multiplication). Since φ is a graded algebra homomorphism:

$$\gamma \circ m_\nu^{\mathcal{O}_2} = m_\nu^{\mathcal{O}_1} \circ (\alpha \otimes \beta).$$

Step 4 (Rank inequality). Since $\alpha \otimes \beta$ is surjective (tensor product of surjections over $\mathbf{C}$ is surjective):

$$\operatorname{rank}(m_\nu^{\mathcal{O}_1}) = \operatorname{rank}\big(m_\nu^{\mathcal{O}_1} \circ (\alpha \otimes \beta)\big) = \operatorname{rank}(\gamma \circ m_\nu^{\mathcal{O}_2}) \leq \operatorname{rank}(m_\nu^{\mathcal{O}_2}),$$

where the last inequality holds because post-composition with any linear map cannot increase rank. Dividing both sides by $\dim L(\nu)$ gives $\mathrm{sr}_{\mathcal{O}_1}^{d_1,d_2}(\lambda, \mu; \nu) \leq \mathrm{sr}_{\mathcal{O}_2}^{d_1,d_2}(\lambda, \mu; \nu)$. □

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